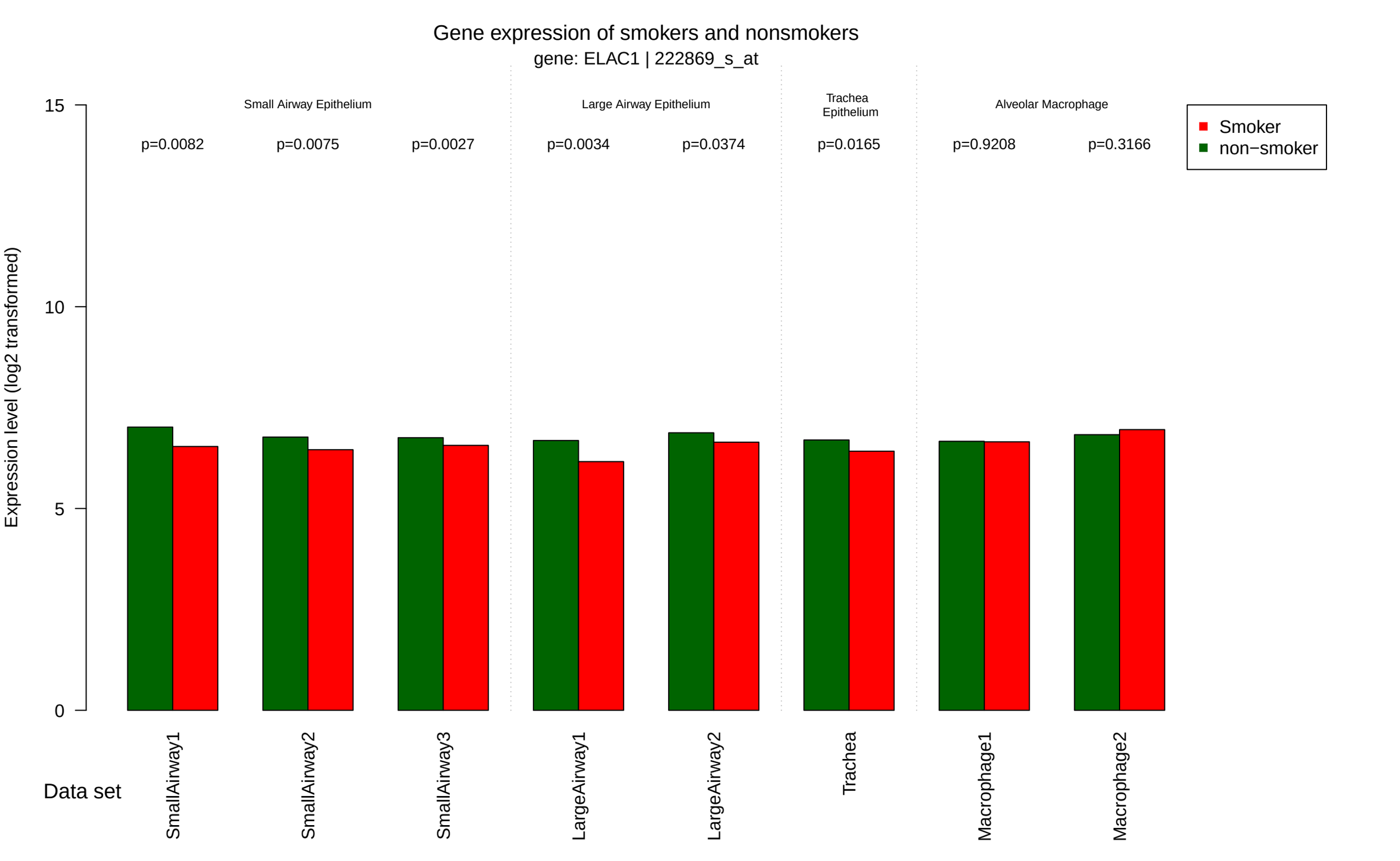
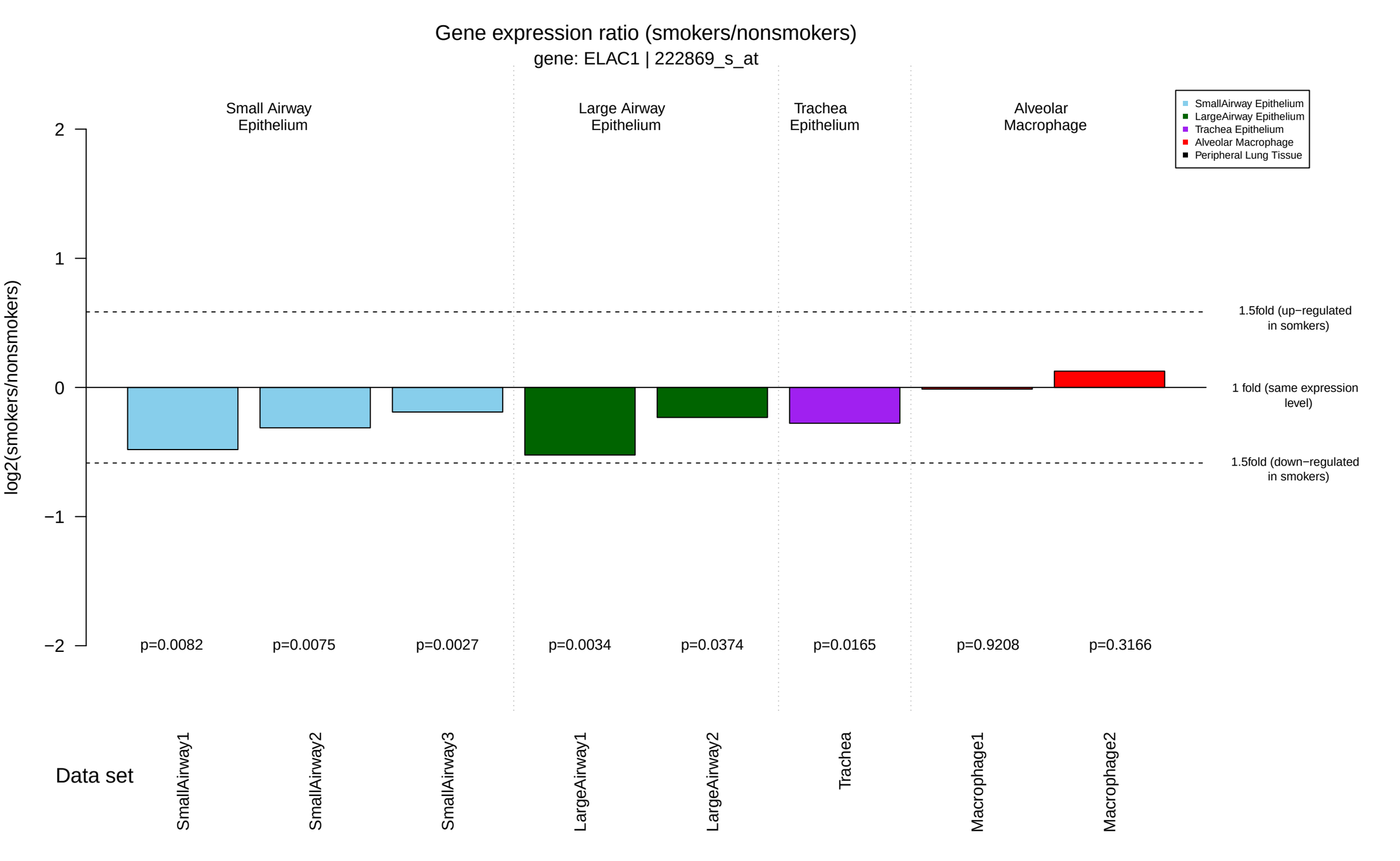
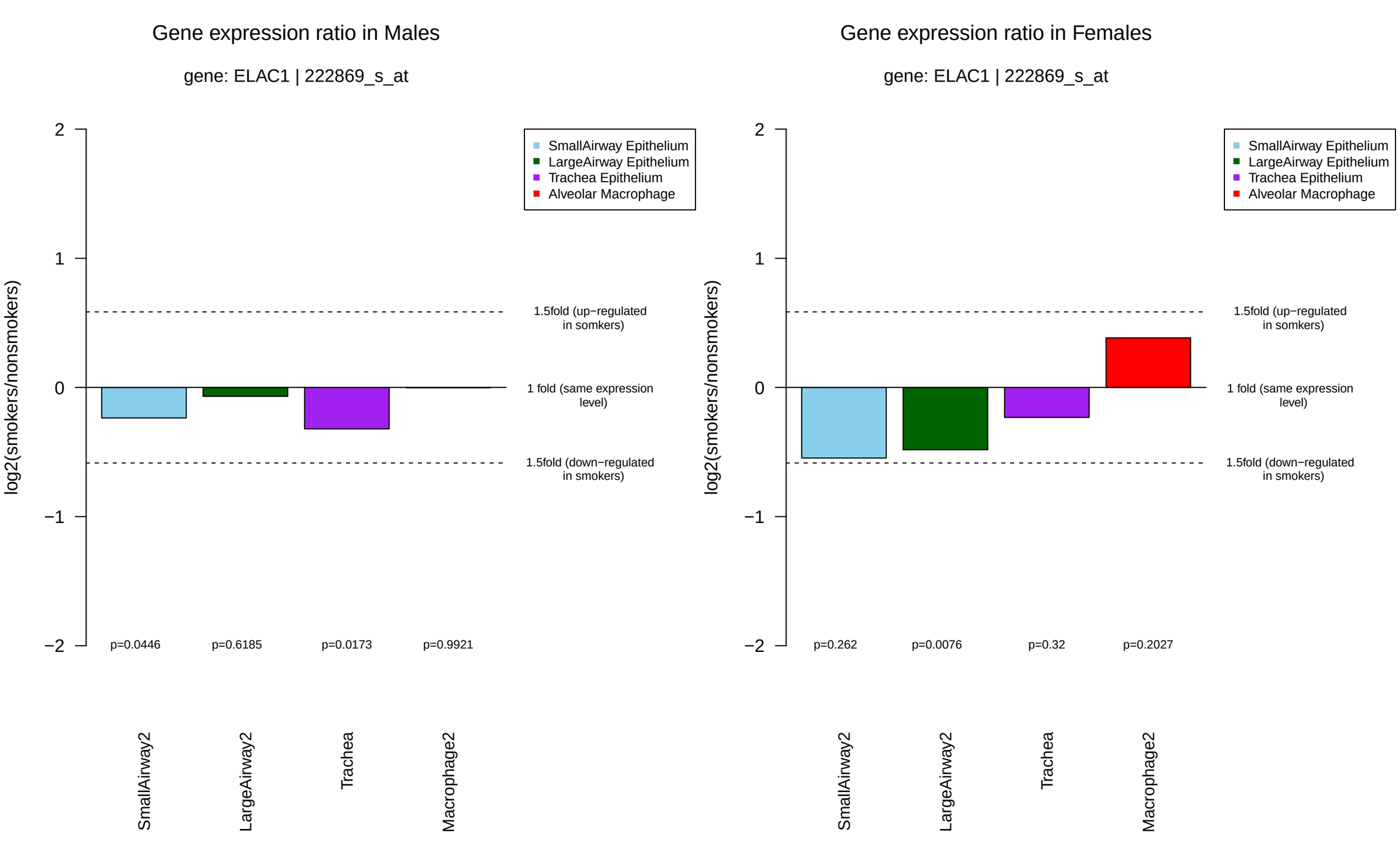
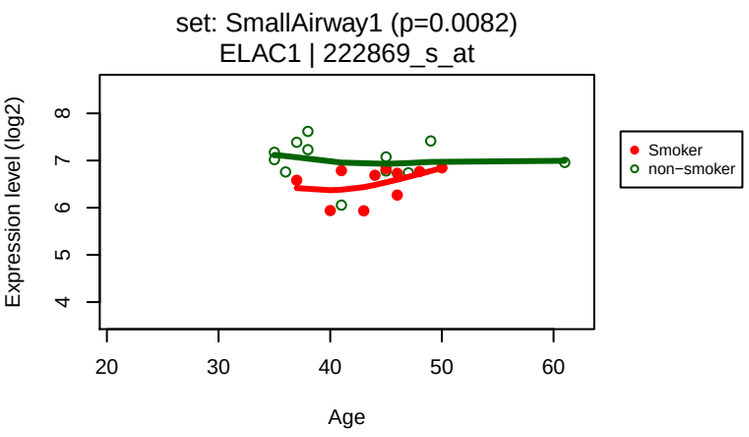
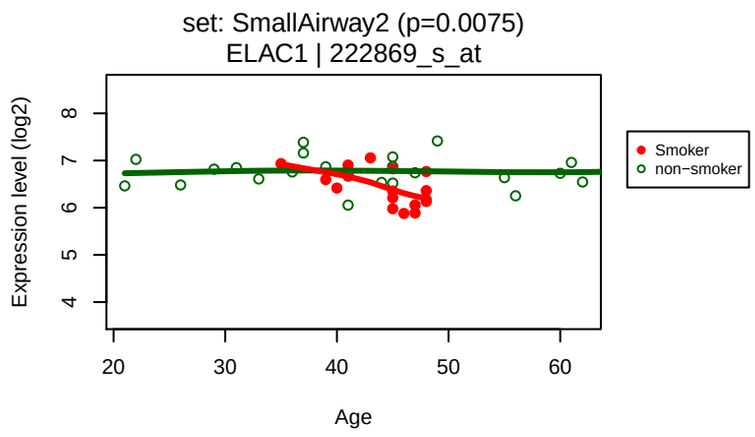
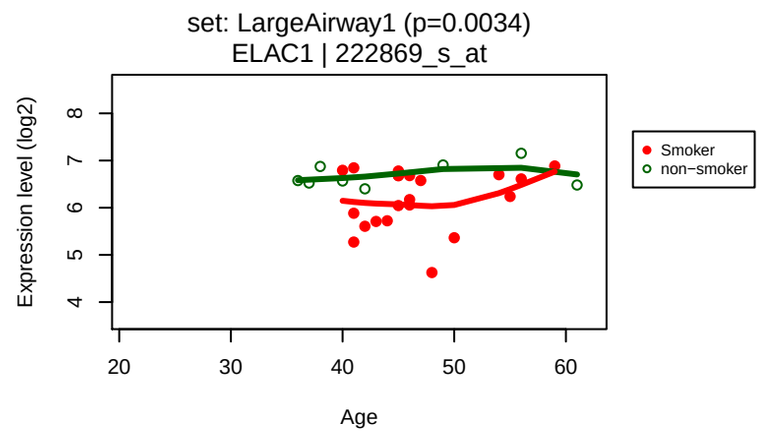

Gene expression of smokers and nonsmokers
gene: ELAC1 | 222869_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0082
p=0.0075
p=0.0027
p=0.0034
p=0.0374
p=0.0165
p=0.9208
p=0.3166
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

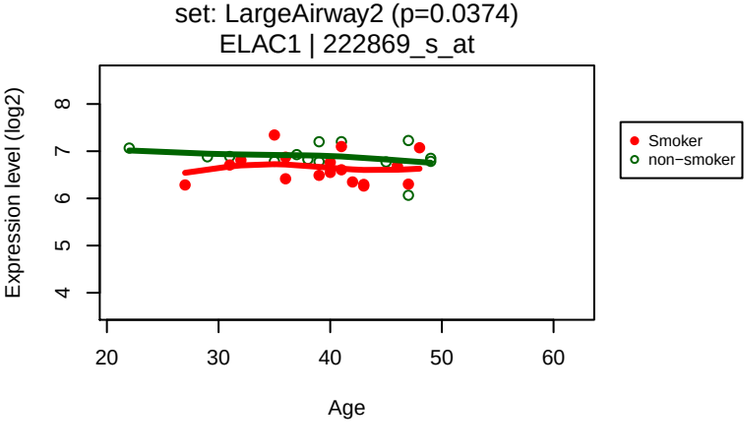

Gene expression ratio (smokers/nonsmokers)
gene: ELAC1 | 222869_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0082
p=0.0075
p=0.0027
p=0.0034
p=0.0374
p=0.0165
p=0.9208
p=0.3166
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

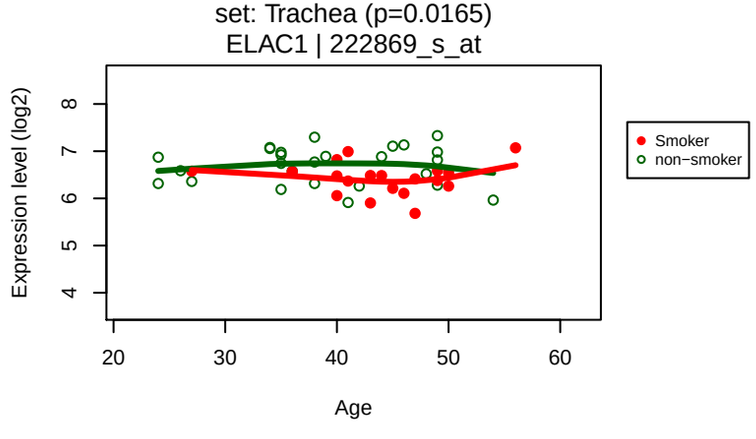

Gene expression ratio in Males
gene: ELAC1 | 222869_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0446
p=0.6185
p=0.0173
p=0.9921
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELAC1 | 222869_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.262
p=0.0076
p=0.32
p=0.2027
SmallAirway2
LargeAirway2
Trachea
Macrophage2

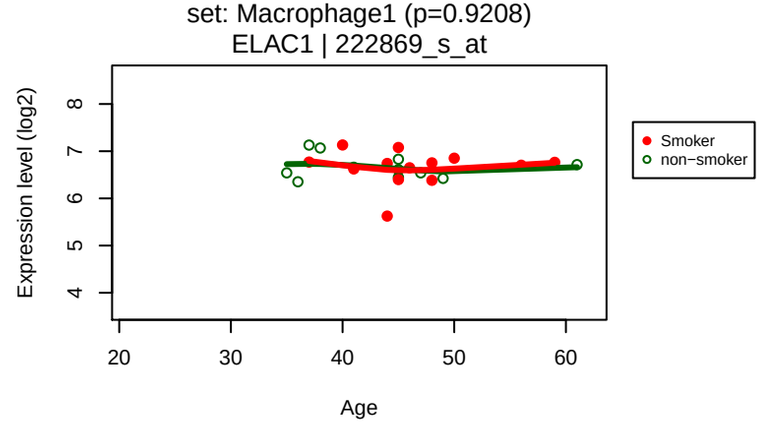

set: SmallAirway1 (p=0.0082)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker

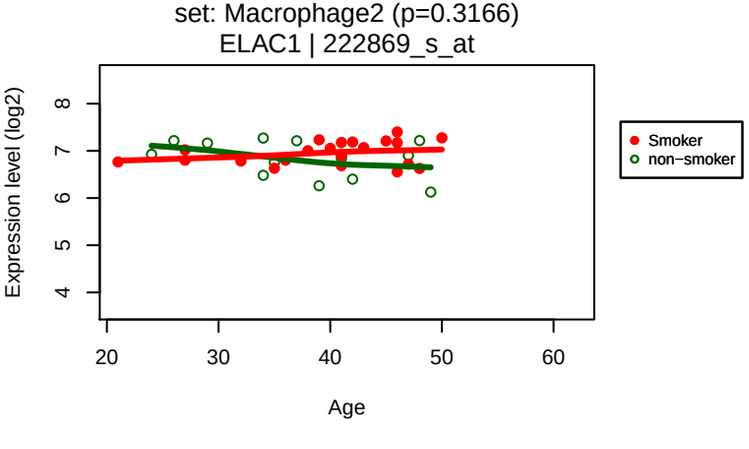

set: SmallAirway2 (p=0.0075)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker

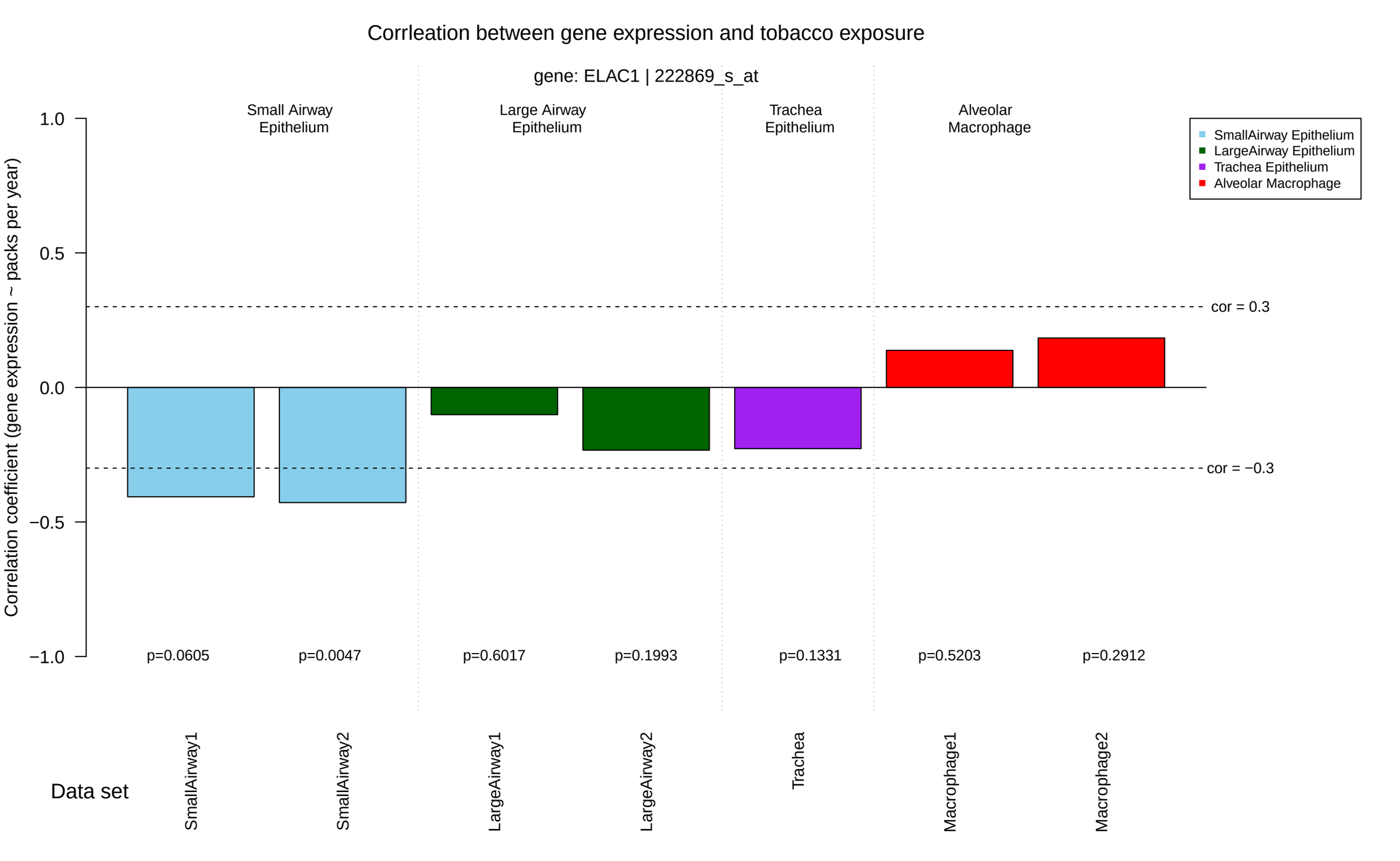

set: LargeAirway1 (p=0.0034)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0374)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0165)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9208)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3166)
ELAC1 | 222869_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELAC1 | 222869_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0605
p=0.0047
p=0.6017
p=0.1993
p=0.1331
p=0.5203
p=0.2912
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2